4.6.

6. Suppose we collect data for a group of students in a statistics class with variables $X_1$ = hours studied, $X_2$ = undergrad GPA, and $Y$ = receive an A. We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6, \hat{\beta}_1 = 0.05, \hat{\beta}_2 = 1$.

(a) Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.

(b) How many hours would the student in part (a) need to study to have a 50 % chance of getting an A in the class?

(a) Gets A

$$P(Y=1) = \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2}}$$

$$= \frac{e^{-6+0.05\times 40 + 1\times 3.5}}{1 + e^{-6+0.05\times 40 + 1\times 3.5}} = \frac{e^{-0.5}}{1+e^{-0.5}} \doteqdot 0.38$$

(b)

$$0.5 = \frac{e^{-6+0.05x_1 + 1\times 3.5}}{1 + e^{-6+0.05x_1 + 3.5}} = \frac{e^{0.05x_1 - 2.5}}{1 + e^{0.05x_1 - 2.5}}$$

$$\frac{1}{2} e^{-6+0.05x_1+3.5} = e^{-6+0.05x_1+3.5}$$

$$\frac{1}{2} \times (-6+0.05x_1+3.5) = -6+0.05x_1+3.5$$

$$\Rightarrow 0.025x_1 = 1.25$$

$$\Rightarrow x_1 = 50$$

9. This problem has to do with *odds*.

(a) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?

(b) Suppose that an individual has a 16 % chance of defaulting on her credit card payment. What are the odds that she will default?

$$\text{Odds} = \frac{p}{1-p} \Rightarrow p = \frac{\text{Odds}}{1+\text{Odds}}$$

(a) $\text{Odds} = \frac{0.37}{1+0.37} = 0.27$

(b) $\text{Odds} = \frac{0.16}{1-0.16} = 0.19$

12. Suppose that you wish to classify an observation $X \in \mathbb{R}$ into `apples` and `oranges`. You fit a logistic regression model and find that

$$\widehat{\Pr}(Y = \texttt{orange}|X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}.$$

Your friend fits a logistic regression model to the same data using the *softmax* formulation in (4.13), and finds that

$$\widehat{\Pr}(Y = \texttt{orange}|X = x) = \frac{\exp(\hat{\alpha}_{\texttt{orange}0} + \hat{\alpha}_{\texttt{orange}1} x)}{\exp(\hat{\alpha}_{\texttt{orange}0} + \hat{\alpha}_{\texttt{orange}1} x) + \exp(\hat{\alpha}_{\texttt{apple}0} + \hat{\alpha}_{\texttt{apple}1} x)}.$$

(a) What is the log odds of `orange` versus `apple` in your model?

(b) What is the log odds of `orange` versus `apple` in your friend's model?

(c) Suppose that in your model, $\hat{\beta}_0 = 2$ and $\hat{\beta}_1 = -1$. What are the coefficient estimates in your friend's model? Be as specific as possible.

(d) Now suppose that you and your friend fit the same two models on a different data set. This time, your friend gets the coefficient estimates $\hat{\alpha}_{\texttt{orange}0} = 1.2$, $\hat{\alpha}_{\texttt{orange}1} = -2$, $\hat{\alpha}_{\texttt{orange}0} = 3$, $\hat{\alpha}_{\texttt{orange}1} = 0.6$. What are the coefficient estimates in your model?

(e) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.

(a) $\text{log-odds} = \log \frac{P(\text{orange}|x)}{P(\text{apple}|x)}$

$P(\text{apple}|x) = 1 - P(\text{orange}|x)$

$\Rightarrow \text{log-odds} = \hat{\beta}_0 + \hat{\beta}_1 x$

(b) $\text{log-odds} = \log \frac{P(\text{orange}|x)}{P(\text{apple}|x)} = (\hat{\alpha}_0^{\text{orange}} - \hat{\alpha}_0^{\text{apple}}) + (\hat{\alpha}_1^{\text{orange}} - \hat{\alpha}_1^{\text{apple}}) x$

(c) $\hat{\beta}_0 = \alpha_0^{\text{orange}} - \alpha_0^{\text{apple}}$, $\hat{\beta}_1 = \alpha_1^{\text{orange}} - \alpha_1^{\text{apple}}$

set the apple coefficients to 0

$\Rightarrow \alpha_0^{\text{apple}} = 0$, $\alpha_1^{\text{apple}} = 0$

$\alpha_0^{\text{orange}} = 2$, $\alpha_1^{\text{orange}} = -1$

(d) $\hat{\beta}_0 = 1.2 - 3 = -1.8$

$\hat{\beta}_1 = -2 - 0.6 = -2.6$

(e) $$P_{\text{your model}}(\text{orange} \mid x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

$$= \frac{e^{(\alpha_0^{\text{orange}} - \alpha_0^{\text{apple}}) + (\alpha_1^{\text{orange}} - \alpha_1^{\text{apple}})x}}{1 + e^{(\alpha_0^{\text{orange}} - \alpha_0^{\text{apple}}) + (\alpha_1^{\text{orange}} - \alpha_1^{\text{apple}})x}}$$

$\Rightarrow$ identical to friend's model

$\Rightarrow$ $P_{\text{you}}(\text{orange} \mid x) = P_{\text{friend}}(\text{orange} \mid x)$

$\Rightarrow$ The predicted class labels will always be identical.